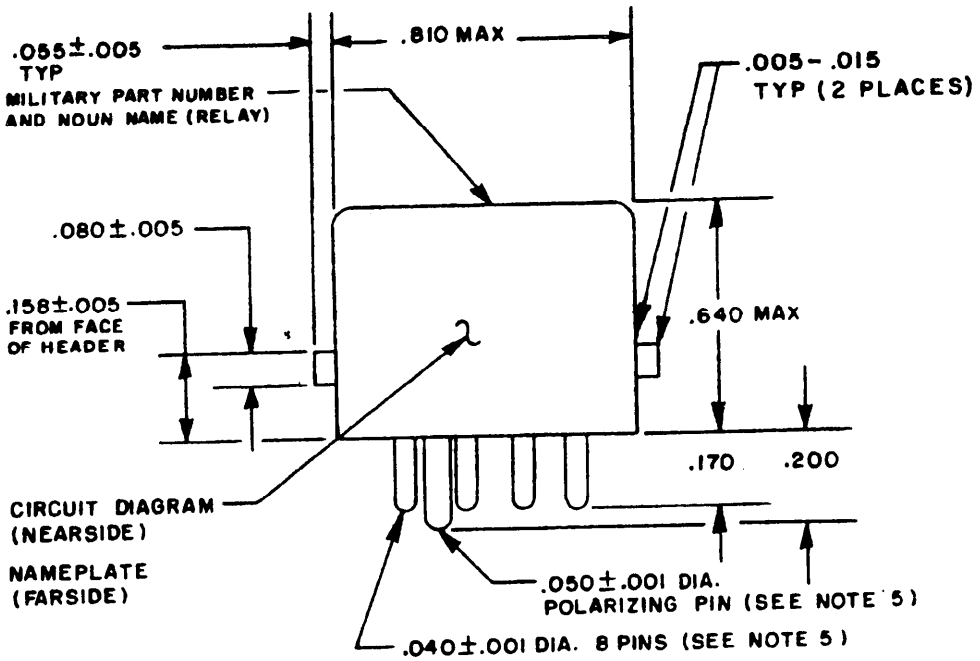
MILITARY SPECIFICATION SHEET

RELAY, SOLID STATE, SENSORS, ESTABLISHED RELIABILITY,
TYPE II (DC VOLTAGE SENSOR), CLASS C, SPST (NO),
250 MILLIAMPERES, HERMETICALLY SEALED (POTTED), PLUG-IN TERMINALS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-R-28894.



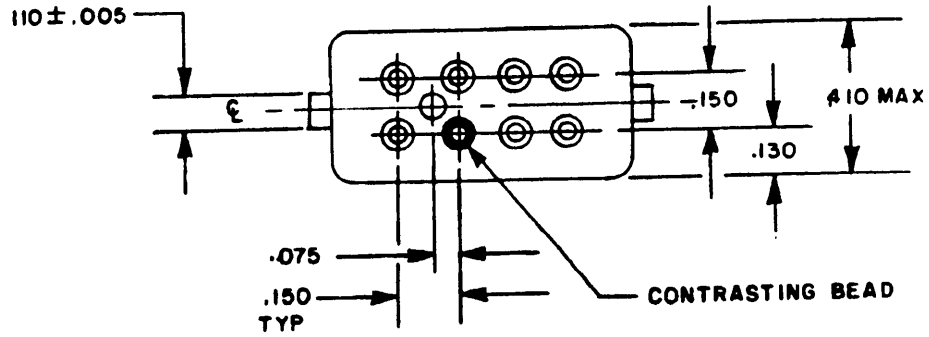
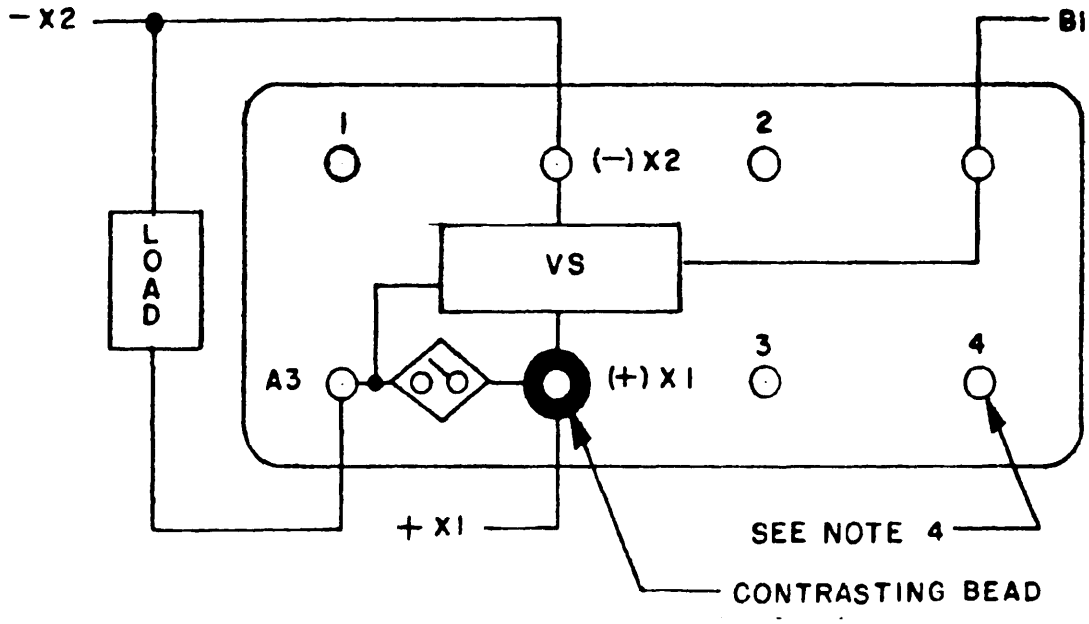


FIGURE 1. Outline dimensions and configuration.

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CIRCUIT DIAGRAM TERMINAL VIEW

INCHES	MM
.001	0.03
.005	0.13
.010	0.25
.015	0.38
.040	1.02
.050	1.27
.055	1.40
.075	1.91
.080	2.03
.110	2.79
.130	3.30
.150	3.81
.158	4.01
.170	4.32
.200	5.08
.410	10.41
.640	16.26
.810	20.57

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Terminal numbers shown above for reference only. Numbers do not appear header.
4. Terminal 1, 2, 3 and 4 are non-functional leads.
5. Finish, all pins, gold per MIL-G-45204, type II, class 1, underplating shall be nickel 50 to 150 microinches thick
6. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.

FIGURE 1. Outline dimensions and configuration - Continued.

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REQUIREMENTS:

Input data: 1/

Input voltage range (B1-X2): 0 to 12 V dc.

Input current (B1-X2): 0.5 mA maximum at 5.0 V.

Operating data:

Operating voltage: 20 to 32 V dc.

Maximum operating current: 50 mA, maximum.

Pickup voltage: 2.0 V dc maximum over temperature range.

Dropout voltage: 1.0 V dc maximum over temperature range.

Turn on time and turn off time: 25 milliseconds, maximum. 2/

Operating mode: When 28 V dc is applied across X1-X2, the output will energize when the sensed voltage is 2.0 V dc, maximum. The output will de-energize when the sensed voltage falls to 0.5 to 1.0 V dc, maximum.

Output data:

Configuration: SPST, NO, Solid state closure to positive.

Contact rating: 250 mA inductive.

ON state voltage drop: 2.0 V dc, maximum.

Electrical data:

Transients: 3/

Positive transients: 80 V in accordance with MIL-STD-704.

Spike transients: ±600 V in accordance with MIL-STD-704.

Dielectric withstanding voltage: 1,000 V rms, 60 Hz 4/.

Insulation resistance @ 500 V dc: 1,000 megohms, minimum 4/.

Dielectric resistance: 3/ 1,000 V rms, 60 Hz at sea level, 350 V rms @ 80 feet.

Timing cycle: Not applicable.

Environmental data:

Ambient temperature (operating or nonoperating): -55°C to +125°C.

1/ Relays are protected against polarity reversal.

2/ Time must be measured at least 25 milliseconds after application of power to X1-X2 and shall be measured as specified in MIL-R-28894 for pickup time and dropout time except contact bounce is not applicable.

3/ Transient and power loss specifications are based on a maximum duty cycle of 1/50.

4/ Dielectric withstanding voltage and insulation resistance are measured between all mutually insulated terminals and between all terminals and case.

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Vibration: 0.06 in. D.A., 10 to 55 Hz; 10G's, 55 to 500 Hz; 20G's, 500 to 3000 Hz.

Shock: 50G's 11 ±1 ms, 1/2 sine, 3 axes.

Acceleration: 20G's in any axis.

Physical data:

Seal: Hermetic, potted.

Dimensions and configuration: See figure 1.

Terminations: See figure 1.

Weight: 16 grams (0.56 ounce) maximum.

Identification marking (full): Applicable.

Qualification inspection: See table I.

TABLE I. Qualification inspection and sample size.

Single submission		Group submission
18 units plus 1 unsealed unit for level L at C = 0 1/	M28894/19-001	18 units plus 1 unsealed unit for level L at C = 0 1/
33 units plus 1 unsealed unit for level M at C = 0 1/		33 units plus 1 unsealed unit for level M at C = 0 1/
Qualification inspection as applicable.		Qualification inspection as applicable.

1/ The number of units required for qualification testing will be increased as required in group V of the qualification inspection table of MIL-R-28894, if the supplier elects to test the number of units permitting one or more failures.

Part number: M28894/19-001 and suffix letter to designate applicable failure rate level. Failure rate level (percent per 1,000 hr.): L, 3.0; M, 1.0; P, 0.1; R, 0.01.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AV
Navy - OS, SH

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5945-0725-19)

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